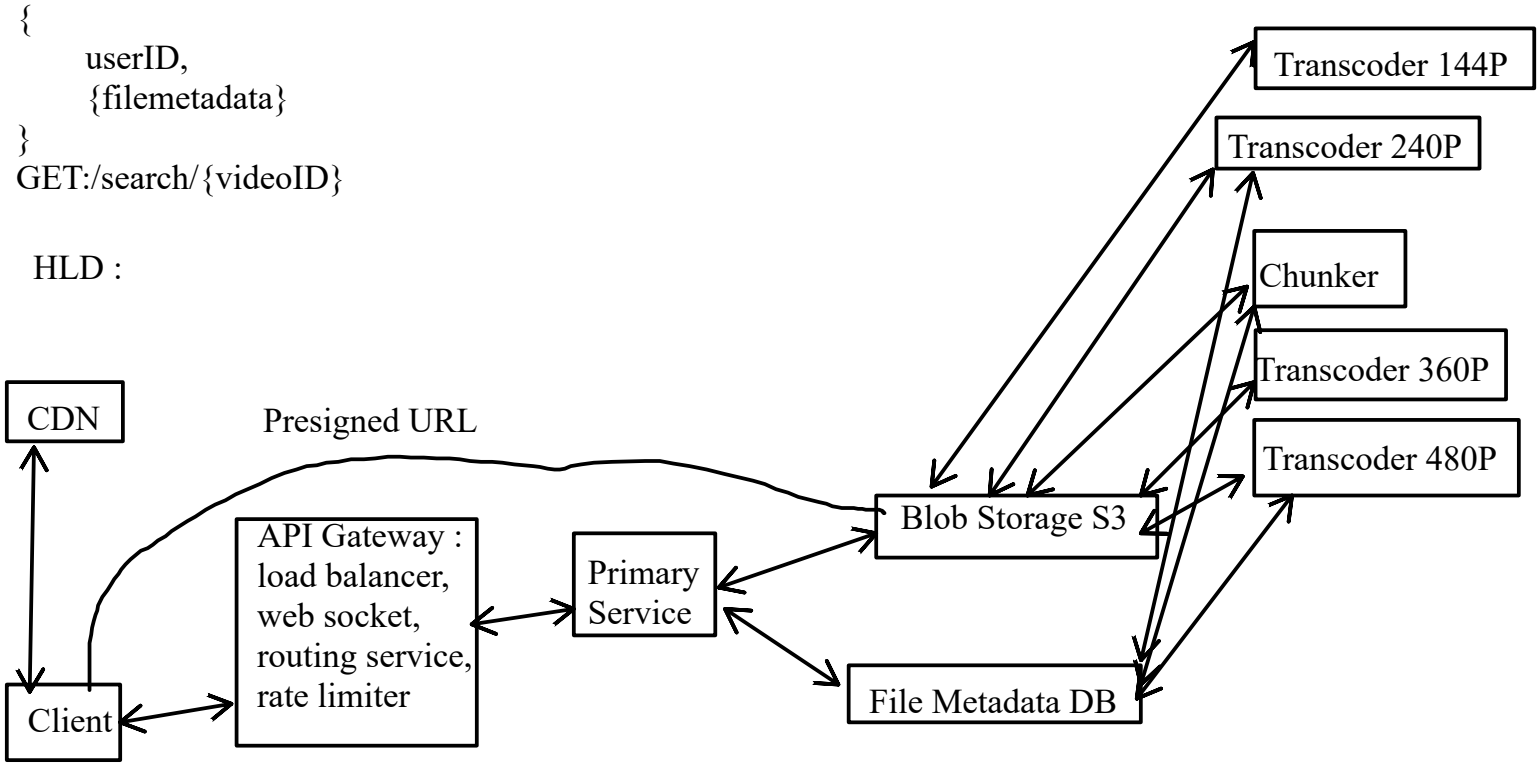
Youtube

Functional requirements:
User should be able to view any video based on his preference,
User should be download any videos based on his preference,
Video resolution should get adjusted based on the Network bandwidth,

Non-Funtional reqruirements:
Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime
i.e Availability>>Consistency)

Core Entities:
user ,
video,
history,
activity

API :
POST:/upload
{
userID,
{filemetadata}
}
GET:/search/{videoID}

HLD :